

HARSHITH VARMA RUDRARAJU

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SUMMARY

AI/ML Engineer • LLM Systems & Agentic AI • Deep Learning & NLP • MLOps & Scalable Inference

EDUCATION

University of Alabama at Birmingham

Master of Science, Data Science

•GPA: 3.66/4.0

•Coursework: Deep Learning, NLP, Probabilistic ML, Advanced Statistics

Aug 2024 - May 2026

Birmingham, USA

EXPERIENCE

AriesView

AI/ML Engineer Intern

Sep 2025 - Jan 2026

Remote - Boston, USA

- Architected a production-grade RAG pipeline using LangChain, FAISS, and OpenAI Ada-002 embeddings, improving semantic retrieval precision by 38% over BM25 baselines across a corpus of 50K+ financial documents.
- Designed and benchmarked a hybrid chunking strategy (semantic + fixed-size) with metadata-aware re-ranking, reducing hallucination rate by 22% as measured against curated ground-truth QA sets.
- Built an OCR-to-LLM extraction pipeline using Tesseract + GPT-4 to convert unstructured investment memos into structured JSON schemas, achieving 97% field-level extraction accuracy on held-out test sets.
- Conducted systematic A/B experiments comparing embedding models (Ada-002 vs. Instructor-XL vs. BGE-large), reducing end-to-end inference latency by 40% while maintaining NDCG@10 retrieval quality.
- Deployed model monitoring via Evidently AI to track embedding drift and answer quality degradation in real time, enabling proactive retraining triggers and maintaining SLA adherence over 3-month production window.

UAB Athletics Department

ML Data Analyst Intern

Sep 2025 - Jan 2026

Birmingham, USA

- Engineered athlete biometric and training-load features using Python (Pandas, SciPy) and PostgreSQL, consolidating 12 disparate data sources into a unified analytics warehouse for downstream ML modeling.
- Trained a gradient-boosted ensemble (XGBoost + LightGBM) on session-RPE, sleep quality, and HRV signals to forecast next-day fatigue levels, achieving RMSE of 0.71 on a 5-point Borg scale holdout.
- Built a time-series attendance prediction model using LSTM networks (PyTorch), reducing unexcused missed sessions by 12% within one semester by enabling 48-hour preemptive coach alerts.
- Developed interactive Power BI dashboards surfacing per-athlete Z-score outliers, training monotony, and recovery readiness indices; adopted by coaching staff to inform weekly periodization decisions.

Inspira AI Innovations Pvt. Ltd.

AI/ML Engineer

Aug 2023 - Jul 2024

Remote-Haryana, India

- Developed and fine-tuned a domain-specific LLM for Vedic Astrology, trained on curated datasets from multiple classical Vedic astrology texts.
- Designed and implemented Retrieval-Augmented Generation (RAG) pipelines to deliver accurate, knowledge-grounded astrological responses.
- Integrated tool calling capabilities, enabling the model to dynamically invoke external functions and APIs for real-time data retrieval and processing.
- Managed and optimized training datasets, ensuring data quality and structure for effective model learning.
- Built and deployed both text-based and voice-based AI response interfaces for the astrology model.

PROJECTS

Token-Efficient Long-Context Summarization via Hierarchical Attention Distillation

Jan 2026 - Present

- Fine-tuned Mistral-7B with LoRA (rank=16) on 18K scientific abstracts from PubMed using a custom hierarchical summarization objective, reducing token cost by 63% versus full-context inference with very less ROUGE-L degradation.
- Implemented sliding-window chunked encoding with cross-chunk attention bridging, enabling 32K-token document summarization on a single A100 GPU without truncation artifacts.
- Evaluated against PEGASUS-X and LongT5 using ROUGE-1/2/L, BERTScore, and human preference ratings; proposed model achieved 2nd-best automated scores while ranking 1st in human coherence ratings (n=42 raters).
- Open-sourced training pipeline and model weights on HuggingFace Hub; documented full reproducibility guide including dataset curation, LoRA config, and evaluation harness.

Multimodal Dietary Assessment: Visual Food Recognition with Portion-Aware Caloric Estimation Aug 2025 - Dec 2025

- Built a two-stage pipeline - YOLOv8-L for real-time food detection (mAP@0.5: 0.82) followed by SAM-based segmentation for

pixel-level portion mask estimation - trained on a custom-annotated dataset of 5,400 meal images.

- Derived volumetric caloric estimates from 2D silhouette area + dish-diameter reference calibration, correcting a 15% systematic bias in high-density foods (nuts, cheese) identified during USDA ground-truth validation.
- Designed a macro-tracking dashboard in Streamlit with longitudinal nutrient trend plots and deficiency flagging logic, achieving 89% user-reported satisfaction in a 3-week pilot with 18 UAB student volunteers.

Contrastive Self-Supervised Pre-training for Low-Resource Tabular Health Classification

Jan 2025 - May 2025

- Pre-trained a tabular encoder using SCARF-style contrastive learning on 95K unlabeled EHR records, then fine-tuned on 4,200 labeled samples for diabetes risk classification - outperforming fully-supervised baselines by 4.1% AUC.
- Systematically compared four classifiers (Logistic Regression, MLP, Kernel SVM, Random Forest) under identical PCA-reduced (95% variance) feature spaces, demonstrating contrastive pre-training consistently boosts minority-class recall.
- Applied SMOTENC for mixed-type feature oversampling, improving diabetic-class F1 from 0.71 to 0.84; final ensemble achieved 87.6% accuracy distinguishing diabetic vs. pre-diabetic with AUC = 0.93.

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, Bash, NumPy, Pandas, SciPy
- **ML & Deep Learning:** PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, LoRA/PEFT, Diffusers
- **LLM & Agentic AI:** LangChain, LlamaIndex, OpenAI API, Anthropic API, LangSmith, FAISS, Chroma, RAG pipelines, multi-agent orchestration
- **Computer Vision:** YOLOv8, Segment Anything Model (SAM), OpenCV, torchvision, image segmentation & object detection
- **NLP:** BERT, BioMedBERT, Instructor-XL, text classification, intent detection, ROUGE, BERTScore
- **MLOps & Cloud:** Docker, AWS, SageMaker, S3, EC2, Evidently AI, MLflow, Git, GitHub Actions
- **Data & Visualization:** PostgreSQL, Power BI, Tableau, Matplotlib, Seaborn, Streamlit, Plotly, A/B testing, EDA, statistical hypothesis testing